

Data Mining:

Concepts and Techniques

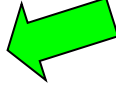
(3rd ed.)

— Chapter 6 —

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Chapter 5: Mining Frequent Patterns, Association and Correlations: Basic Concepts and Methods

- Basic Concepts 
- Frequent Itemset Mining Methods
- Which Patterns Are Interesting?—Pattern Evaluation Methods
- Summary

What Is Frequent Pattern Analysis?

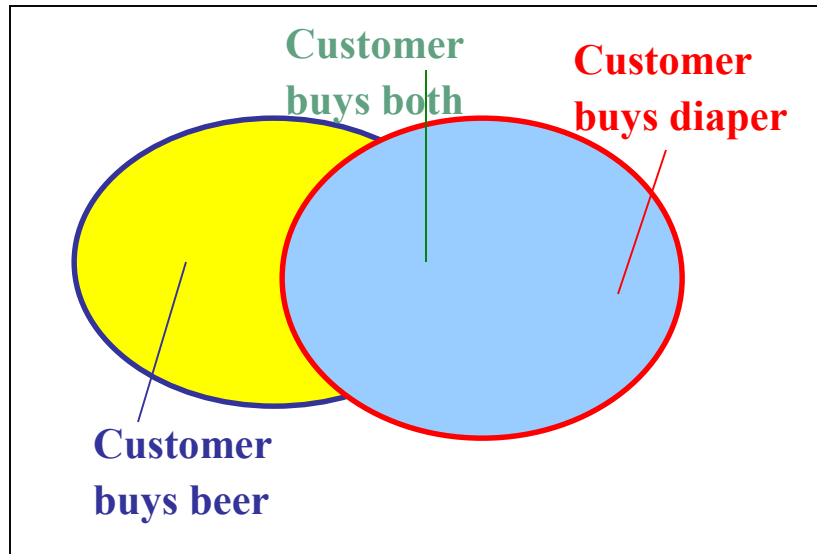
- **Frequent pattern**: a pattern (a set of items, subsequences, substructures, etc.) that occurs frequently in a data set
- Motivation: Finding inherent regularities in data
 - What products were often purchased together?— Beer and diapers?!
 - What are the subsequent purchases after buying a PC?
 - What kinds of DNA are sensitive to this new drug?
 - Can we automatically classify web documents?
- Applications
 - Basket data analysis, cross-marketing, catalog design, sale campaign analysis, Web log (click stream) analysis, and DNA sequence analysis.

Why Is Freq. Pattern Mining Important?

- Freq. pattern: An intrinsic and important property of datasets
- Foundation for many essential data mining tasks
 - Association, correlation, and causality analysis
 - Sequential, structural (e.g., sub-graph) patterns
 - Pattern analysis in spatiotemporal, multimedia, time-series, and stream data
 - Classification: discriminative, frequent pattern analysis
 - Cluster analysis: frequent pattern-based clustering
 - Data warehousing: iceberg cube and cube-gradient
 - Semantic data compression: fascicles
 - Broad applications

Basic Concepts: Frequent Patterns

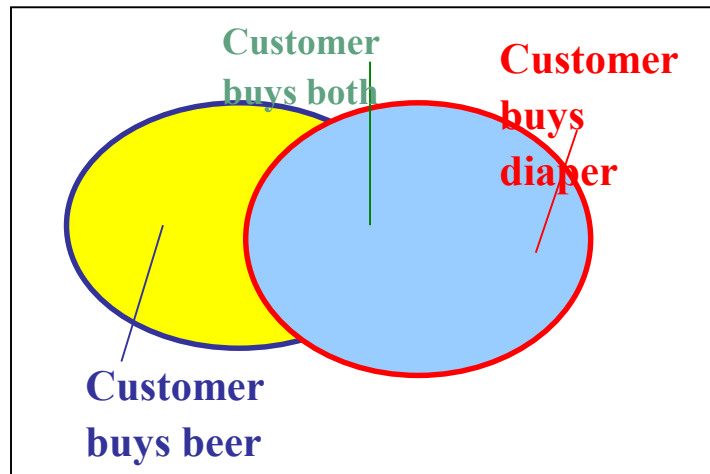
Tid	Items bought
10	Beer, Nuts, Diaper
20	Beer, Coffee, Diaper
30	Beer, Diaper, Eggs
40	Nuts, Eggs, Milk
50	Nuts, Coffee, Diaper, Eggs, Milk



- **itemset**: A set of one or more items
- **k-itemset** $X = \{x_1, \dots, x_k\}$
- **(absolute) support**, or, **support count** of X : Frequency or occurrence of an itemset X
- **(relative) support**, s , is the fraction of transactions that contains X (i.e., the **probability** that a transaction contains X)
- An itemset X is **frequent** if X 's support is no less than a **minsup** threshold

Basic Concepts: Association Rules

Tid	Items bought
10	Beer, Nuts, Diaper
20	Beer, Coffee, Diaper
30	Beer, Diaper, Eggs
40	Nuts, Eggs, Milk
50	Nuts, Coffee, Diaper, Eggs, Milk



- Find all the rules $X \rightarrow Y$ with minimum support and confidence
 - **support**, s , **probability** that a transaction contains $X \cup Y$
 - **confidence**, c , **conditional probability** that a transaction having X also contains Y

Let $minsup = 50\%$, $minconf = 50\%$

Freq. Pat.: Beer:3, Nuts:3, Diaper:4, Eggs:3,
{Beer, Diaper}:3

- Association rules: (many more!)
 - $Beer \rightarrow Diaper$ (60%, 100%)
 - $Diaper \rightarrow Beer$ (60%, 75%)

Closed Patterns and Max-Patterns

- A long pattern contains a combinatorial number of sub-patterns, e.g., $\{a_1, \dots, a_{100}\}$ contains $\binom{100}{1} + \binom{100}{2} + \dots + \binom{100}{100} = 2^{100} - 1 = 1.27 \times 10^{30}$ sub-patterns!
- Solution: Mine *closed patterns* and *max-patterns* instead
- An itemset X is **closed** if X is *frequent* and there exists *no super-pattern* $Y \supset X$, with the same support as X An itemset X is a **max-pattern** if X is frequent and there exists no frequent super-pattern $Y \supset X$
- Closed pattern is a lossless compression of freq. patterns
 - Reducing the # of patterns and rules

Closed Patterns and Max-Patterns

- Exercise. $DB = \{ \langle a_1, \dots, a_{100} \rangle, \langle a_1, \dots, a_{50} \rangle \}$
 - $Min_sup = 1$.
- What is the set of **closed itemset**?
 - $\langle a_1, \dots, a_{100} \rangle: 1$
 - $\langle a_1, \dots, a_{50} \rangle: 2$
- What is the set of **max-pattern**?
 - $\langle a_1, \dots, a_{100} \rangle: 1$
- What is the set of **all patterns**?
 - !!

Example:

Closed Patterns and Max-Patterns

$\{A\} = 4$; not closed due to $\{A,E\}$
 $\{B\} = 2$; not frequent $\Rightarrow$ ignore
 $\{C\} = 5$; not closed due to $\{C,E\}$
 $\{D\} = 4$; not closed: not maximal due to e.g.
 $\{D,E\}$
 $\{E\} = 6$; closed, but not maximal due to e.g.
 $\{D,E\}$

$\{A,B\} = 1$; not frequent $\Rightarrow$ ignore
 $\{A,C\} = 3$; not closed due to $\{A,C,E\}$
 $\{A,D\} = 3$; not closed due to $\{A,D,E\}$
 $\{A,E\} = 4$; closed, but not maximal due to
 $\{A,D,E\}$
 $\{B,C\} = 2$; not frequent $\Rightarrow$ ignore
 $\{B,D\} = 0$; not frequent $\Rightarrow$ ignore
 $\{B,E\} = 2$; not frequent $\Rightarrow$ ignore
 $\{C,D\} = 3$; not closed due to $\{C,D,E\}$
 $\{C,E\} = 5$; closed, but not maximal due to
 $\{C,D,E\}$
 $\{D,E\} = 4$; closed, but not maximal due to
 $\{A,D,E\}$

My dataset:

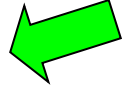
1: A,B,C,E
2: A,C,D,E
3: B,C,E
4: A,C,D,E
5: C,D,E
6: A,D,E

minsupp=50%

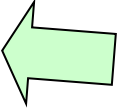
$\{A,B,C\} = 1$; not frequent $\Rightarrow$ ignore
 $\{A,B,D\} = 0$; not frequent $\Rightarrow$ ignore
 $\{A,B,E\} = 1$; not frequent $\Rightarrow$ ignore
 $\{A,C,D\} = 2$; not frequent $\Rightarrow$ ignore
 $\{A,C,E\} = 3$; maximal frequent
 $\{A,D,E\} = 3$; maximal frequent
 $\{B,C,D\} = 0$; not frequent $\Rightarrow$ ignore
 $\{B,C,E\} = 2$; not frequent $\Rightarrow$ ignore
 $\{C,D,E\} = 3$; maximal frequent

$\{A,B,C,D\} = 0$; not frequent $\Rightarrow$ ignore
 $\{A,B,C,E\} = 1$; not frequent $\Rightarrow$ ignore
 $\{B,C,D,E\} = 0$; not frequent $\Rightarrow$ ignore

Chapter 5: Mining Frequent Patterns, Association and Correlations: Basic Concepts and Methods

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Scalable Frequent Itemset Mining Methods

- Apriori: A Candidate Generation-and-Test Approach 
- Improving the Efficiency of Apriori
- FPGrowth: A Frequent Pattern-Growth Approach
- ECLAT: Frequent Pattern Mining with Vertical Data Format

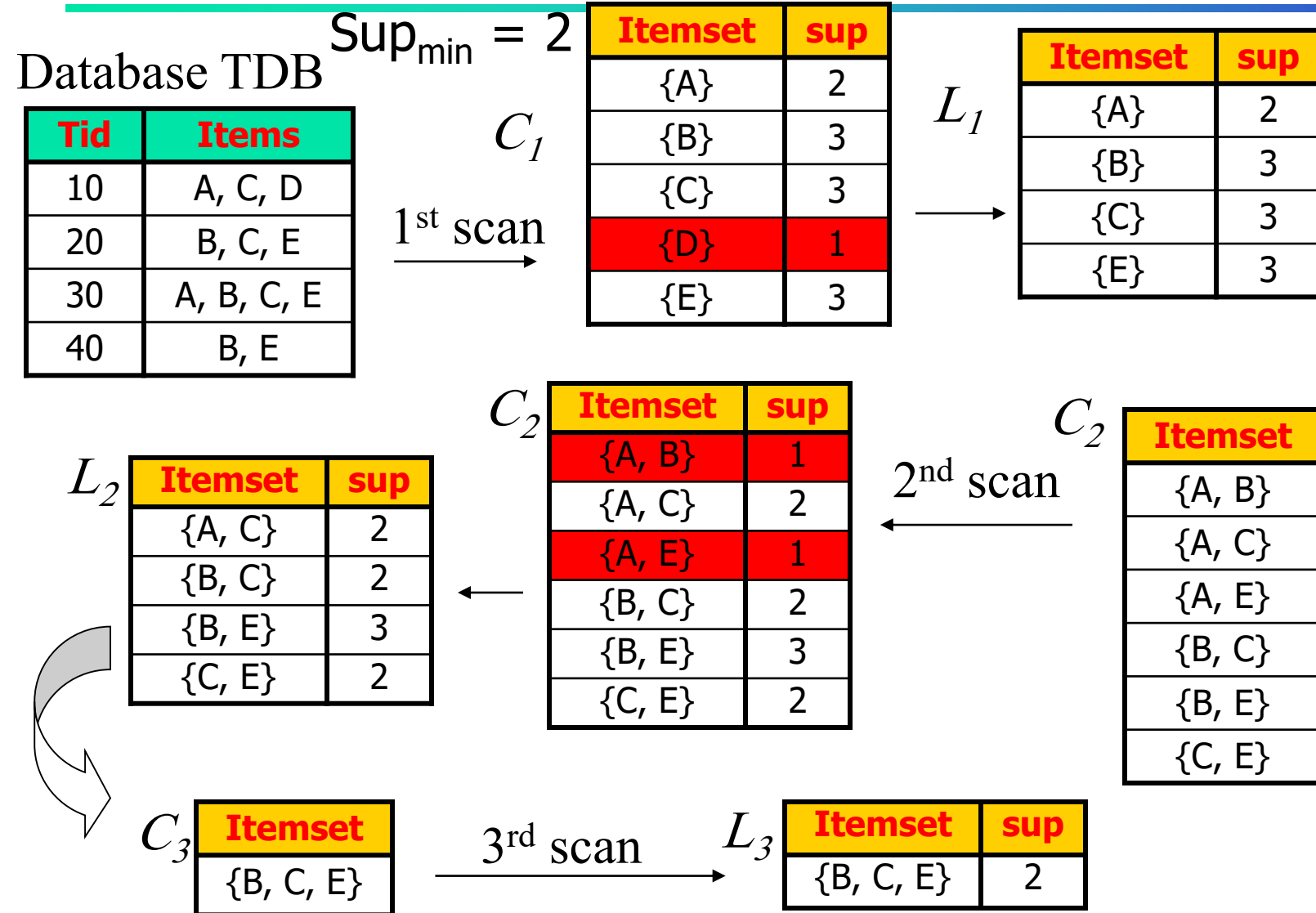
Apriori: A Candidate Generation & Test Approach

- Apriori pruning principle: If there is **any** itemset which is infrequent, its superset should not be generated/tested!

Method:

- Initially, scan DB once to get frequent 1-itemset
- **Generate** length $(k+1)$ **candidate** itemsets from length k **frequent** itemsets
- **Test** the candidates against DB
- Terminate when no frequent or candidate set can be generated

The Apriori Algorithm—An Example



The Apriori Algorithm—An Example

Table 6.1 Transactional Data for an *AlIElectronics* Branch

TID	List of item_IDs
T100	I1, I2, I5
T200	I2, I4
T300	I2, I3
T400	I1, I2, I4
T500	I1, I3
T600	I2, I3
T700	I1, I3
T800	I1, I2, I3, I5
T900	I1, I2, I3

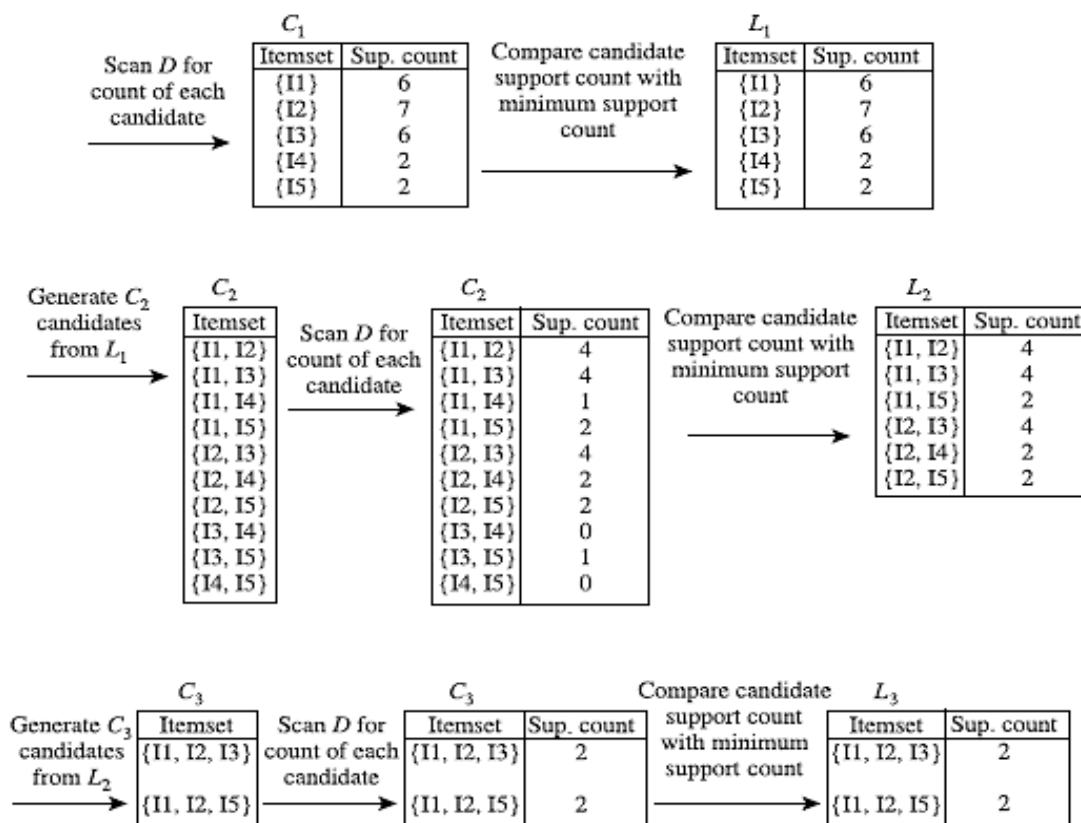


Figure 6.2 Generation of the candidate itemsets and frequent itemsets, where the minimum support count is 2.

The Apriori Algorithm (Pseudo-Code)

C_k : Candidate itemset of size k

L_k : frequent itemset of size k

$L_1 = \{\text{frequent items}\};$

for ($k = 1; L_k \neq \emptyset; k++$) **do begin**

C_{k+1} = candidates generated from L_k ;

for each transaction t in database do

increment the count of all candidates in C_{k+1} that
are contained in t

L_{k+1} = candidates in C_{k+1} with min_support

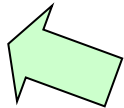
end

return $\cup_k L_k$;

Implementation of Apriori

- How to generate candidates?
 - Step 1: self-joining L_k
 - Step 2: pruning
- Example of Candidate-generation
 - $L_3 = \{abc, abd, acd, ace, bcd\}$
 - Self-joining: $L_3 * L_3$
 - $abcd$ from abc and abd
 - $acde$ from acd and ace
 - Pruning:
 - $acde$ is removed because ade is not in L_3
 - $C_4 = \{abcd\}$

Scalable Frequent Itemset Mining Methods

- Apriori: A Candidate Generation-and-Test Approach
- Improving the Efficiency of Apriori
- FPGrowth: A Frequent Pattern-Growth Approach 
- ECLAT: Frequent Pattern Mining with Vertical Data Format
- Mining Close Frequent Patterns and Maxpatterns

Pattern-Growth Approach: Mining Frequent Patterns Without Candidate Generation

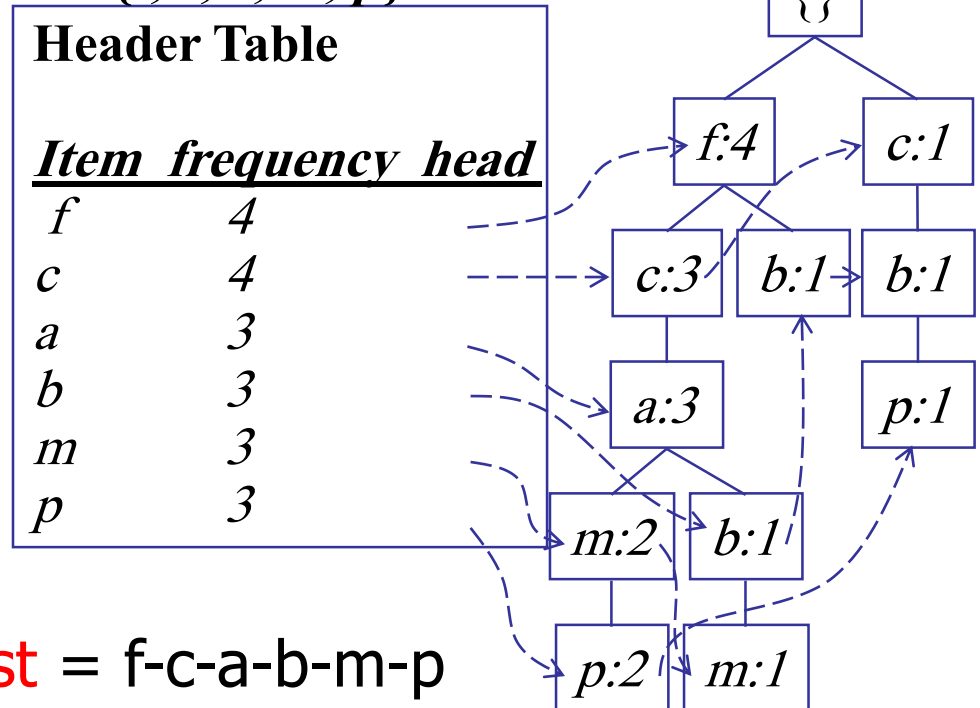
- Bottlenecks of the Apriori approach
 - Breadth-first (i.e., level-wise) search
 - Candidate generation and test
 - Often generates a huge number of candidates
- The FPGrowth Approach (J. Han, J. Pei, and Y. Yin, SIGMOD' 00)
 - Depth-first search
 - Avoid explicit candidate generation
- Major philosophy: Grow long patterns from short ones using local frequent items only
 - "abc" is a frequent pattern
 - Get all transactions having "abc", i.e., project DB on abc: DB|abc
 - "d" is a local frequent item in DB|abc → abcd is a frequent pattern

Construct FP-tree from a Transaction Database

<i>TID</i>	<i>Items bought</i>	<i>(ordered) frequent items</i>
100	{ <i>f, a, c, d, g, i, m, p</i> }	{ <i>f, c, a, m, p</i> }
200	{ <i>a, b, c, f, l, m, o</i> }	{ <i>f, c, a, b, m</i> }
300	{ <i>b, f, h, j, o, w</i> }	{ <i>f, b</i> }
400	{ <i>b, c, k, s, p</i> }	{ <i>c, b, p</i> }
500	{ <i>a, f, c, e, l, p, m, n</i> }	{ <i>f, c, a, m, p</i> }

min_support = 3

1. Scan DB once, find frequent 1-itemset (single item pattern)
2. Sort frequent items in frequency descending order, f-list
3. Scan DB again, construct FP-tree



Partition Patterns and Databases

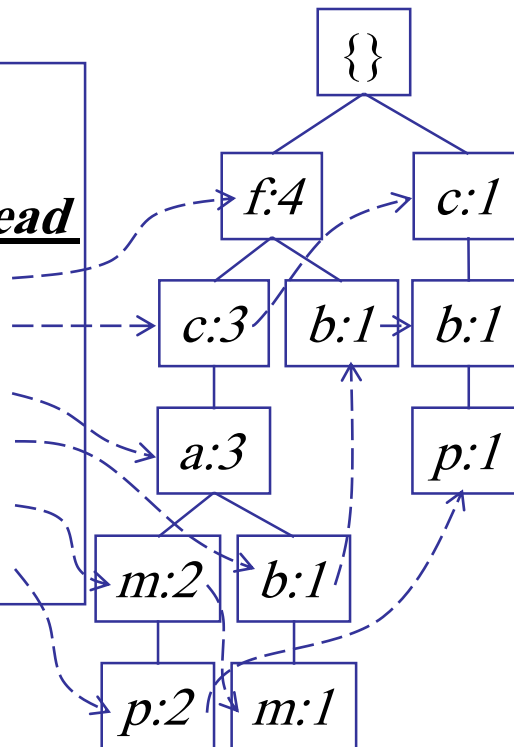
- Frequent patterns can be partitioned into subsets according to f-list
 - F-list = f-c-a-b-m-p
 - Patterns containing p
 - Patterns having m but no p
 - ...
 - Patterns having c but no a nor b, m, p
 - Pattern f
- Completeness and non-redundancy

Find Patterns Having P From P-conditional Database

- Starting at the frequent item header table in the FP-tree
- Traverse the FP-tree by following the link of each frequent item p
- Accumulate all of *transformed prefix paths* of item p to form p 's conditional pattern base

Header Table

<i>Item</i>	<i>frequency</i>	<i>head</i>
<i>f</i>	4	
<i>c</i>	4	
<i>a</i>	3	
<i>b</i>	3	
<i>m</i>	3	
<i>p</i>	3	

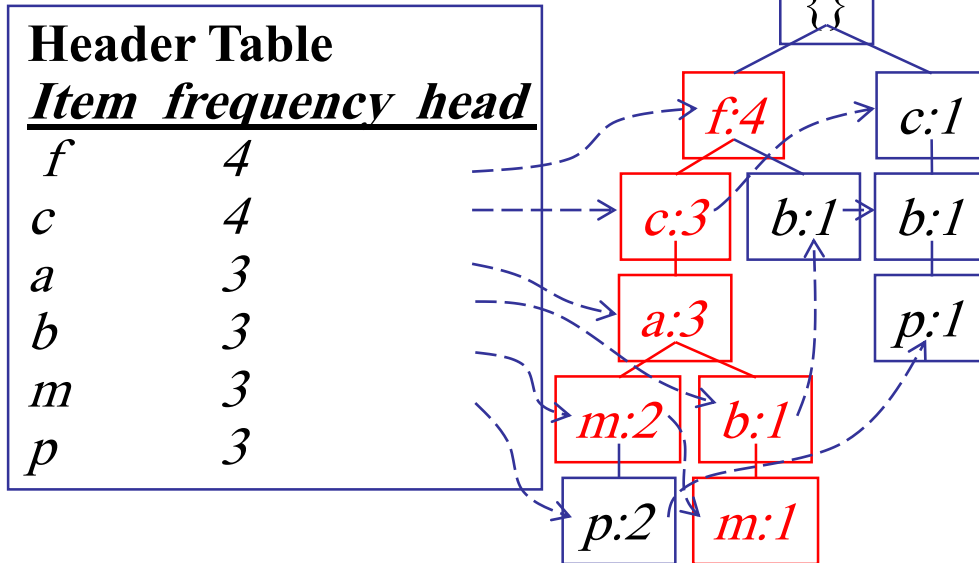


Conditional pattern bases

<i>item</i>	<i>cond. pattern base</i>
<i>c</i>	<i>f:3</i>
<i>a</i>	<i>fc:3</i>
<i>b</i>	<i>fca:1, f:1, c:1</i>
<i>m</i>	<i>fca:2, fcab:1</i>
<i>p</i>	<i>fcam:2, cb:1</i>

From Conditional Pattern-bases to Conditional FP-trees

- For each pattern-base
 - Accumulate the count for each item in the base
 - Construct the FP-tree for the frequent items of the pattern base



m-conditional pattern base:
fca:2, fcab:1



{}

|
f:3

|
c:3

|
a:3

m-conditional FP-tree

All frequent patterns relate to *m*
m,
fm, cm, am,
fc_m, fa_m, ca_m,
fc_ma_m